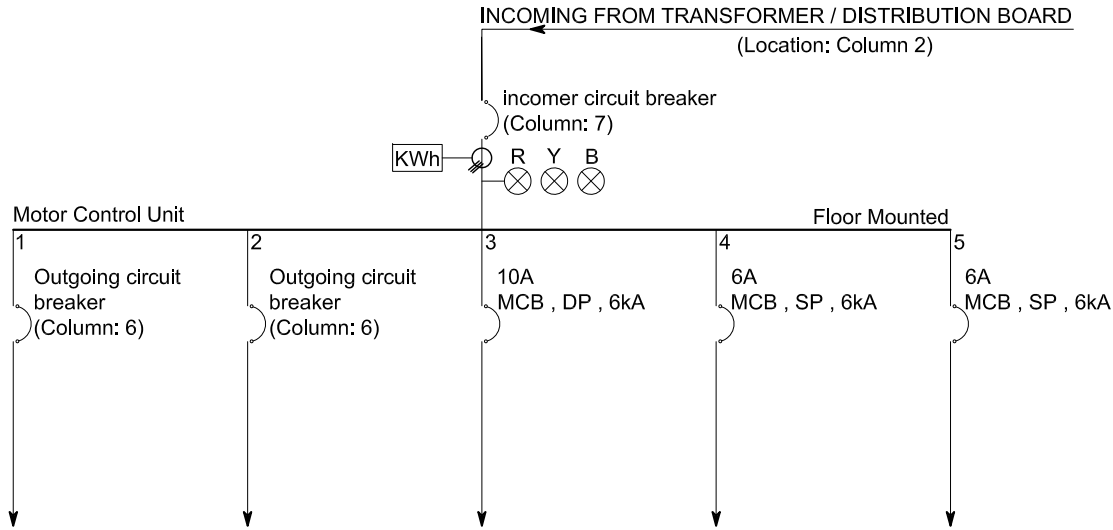


Note.
1- PLEASE REFER TO BELOW TABLE FOR
THE COLUMN NUMBERS MENTIONED IN
THIS DRAWING



Equipment Description	VENTILLATION FAN 1	VENTILATION FAN 2 (if Applicable)	I&C System	Lights	Spare/misc.
Equipment Rating (kW))	COLUMN: 3	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Drop Structure Odor Control System						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir (D1)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
2	Infantry (D2)	4	1	DOL	MCB -TP, 10A-16kA	MCCB-TP, 20A - 25kA
3	Bird Market (D3)	7.5	1	DOL	MCB -TP, 20A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
4	Gov House (D4)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
5	Lakshami Chowk (D5)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
6	Line-A Qurtaba (D6)	15	2	DOL	MCB -TP, 40A-16kA	MCCB-TP, 50A - 25kA
7	Grumangat(D7)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
8	Q Block (D8)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
9	G Block (D10)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		2.2	1	DOL	MCB -TP, 6A-16kA	
10	Zafar Ali Rd (D11)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
11	Ichra (D12)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
12	Shama (D14)	5.5	1	DOL	MCB -TP, 16A-16kA	MCCB-TP, 32A - 25kA
		1.5	1	DOL	MCB -TP, 6A-16kA	
13	Line-B Qurtaba (D16)	11	1	ASD	MCB -TP, 32A-16kA	MCCB-TP, 50A - 25kA
		7.5	1	DOL	MCB -TP, 20A-16kA	
14	Lyton (D17)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
15	PIA (D18)	1.5	1	DOL	MCB -TP, 6A-16kA	MCCB-TP, 20A - 25kA
16	Poonch (D19)	4	1	DOL	MCB -TP, 16-16kA	MCCB-TP, 20A - 25kA

SYMBOL	LEGENDS
	DESCRIPTION
	CIRCUIT BREAKER
MCCB	MOLDED CASE CIRCUIT BREAKER
MCB	MINIATURE CIRCUIT BREAKER
FP	FOUR POLE
TP	THREE POLE
DP	DOUBLE POLE
SP	SINGLE POLE
	EARTH SWITCH
	AMMETER SELECTOR SWITCH
	VOLTMETER SELECTOR SWITCH
	DIGITAL POWER METER
	CURRENT TRANSFORMER
	INDICATION LIGHT

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



A Joint Venture

04		
03		
02		
01		
REV.	DATE	DESCRIPTION

DRAWN
CHECKED BY
RECOMMENDED
CHD./VER.
APPROVED

MUHAMMAD ARSLAN

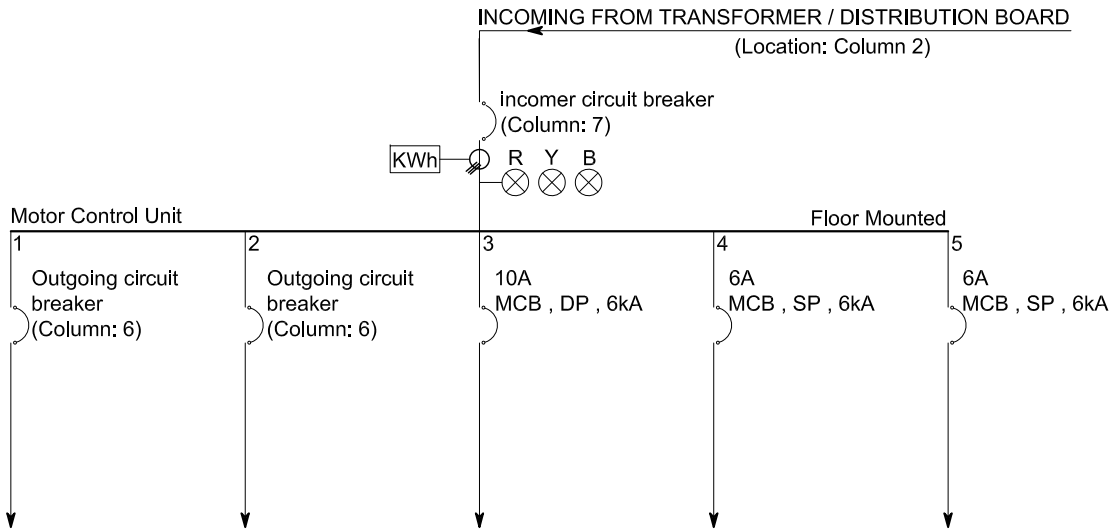
SHAHZAIB SHABBIR

ODOR CONTROL SYSTEM (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-151

Note.
1- PLEASE REFER TO BELOW TABLE FOR
THE COLUMN NUMBERS MENTIONED IN
THIS DRAWING



Equipment Description	SCREEN	CONVEYOR (if Applicable)	I&C System	Lights	Spare/misc.
Equipment Rating (kW))	COLUMN: 3	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Screens Area						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	Main Mir	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
2	infantry	3	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 16A-16kA
3	Bird Market	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
4	Larech Colony	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
5	Ghazi Mohallah	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
6	Mohammad Nagar	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
7	Lakshmi Chowk	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Lakshmi Chowk Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
8	Gov House	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
9	Janki Devi	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
10	Grumangat	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
11	Q Block	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
12	Center Point	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
13	G Block	1.5	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
14	Shareef Colony	0.55	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
15	Zafar Ali Rd	0.37	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
16	Shadman	2.2	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
17	Shama	0.75	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
18	Ichra	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
19	Rasool Park	2.2	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Rasool Park Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	
20	Lyton /Lake Rd	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
21	PIA	1.1	1	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
22	Poonch	1.5	2	DOL	MCB -TP, 6A-16kA	MCB -TP, 16A-16kA
	Poonch Belt Conveyor	0.75	1	DOL	MCB -TP, 6A-16kA	

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



A Joint Venture

04		
03		
02		
01		
REV.	DATE	DESCRIPTION

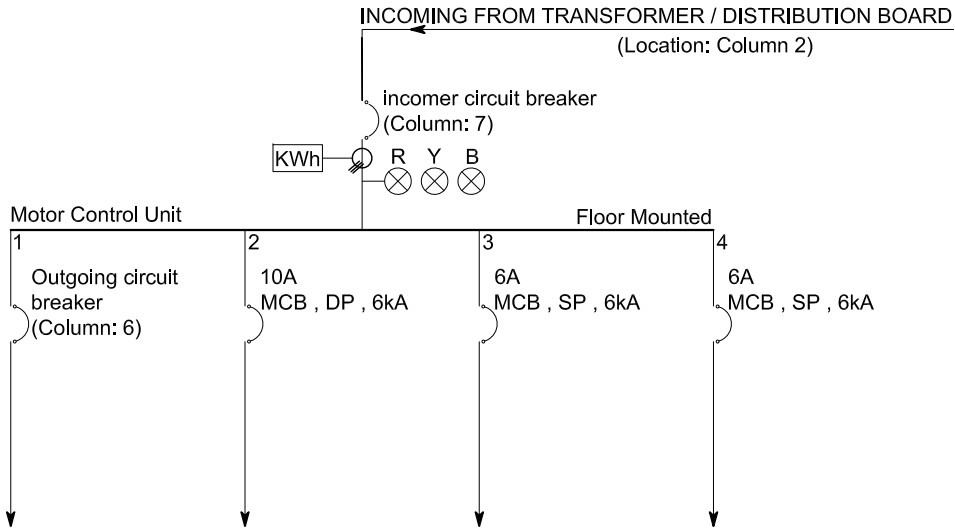
DRAWN	MUHAMMAD ARSLAN
CHECKED BY	SHAHZAIB SHABBIR
RECOMMENDED	
CHD./VER.	
APPROVED	

SCREENS (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-152

Note.
1- PLEASE REFER TO BELOW TABLE FOR THE COLUMN NUMBERS MENTIONED IN THIS DRAWING



Equipment Description	SCREEN	I&C System	Lights	Spare/misc.
Equipment Rating (kW)	COLUMN: 3	--	--	--
Quantity (Nos.)	COLUMN: 4	02	04	04
Starter Type	COLUMN: 5	--	--	--

Column:1	Column:2	Column:3	Column:4	Column:5	Column:6	Column:7
Ventillation System						
Sr.No.	Location	installed Equipment Details				Incoming Circuit Breaker
		Equipment Rating	Installed Qty.	Starter Type	Feeder Circuit Breaker	
		(kW)	(Nos.)			
1	E1	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
2	E2	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
3	E3	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
4	E4	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
5	E5	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
6	E6	2.2	2	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
7	E7	3	1	DOL	MCB -TP, 10A-16kA	MCB -TP, 32A-16kA
8	E8	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
9	E9	5.5	1	DOL	MCB -TP, 16A-16kA	MCB -TP, 32A-16kA
10	E10	7.5	1	DOL	MCB -TP, 20A-16kA	MCB -TP, 40A-16kA
11	E11	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
12	E12	22	1	ASD	MCB -TP, 50A-16kA	MCB -TP, 63A-16kA
13	E13	15	1	ASD	MCB -TP, 40A-16kA	MCB -TP, 63A-16kA

	LEGENDS
SYMBOL	DESCRIPTION
	CIRCUIT BREAKER
MCCB	MOLDED CASE CIRCUIT BREAKER
MCB	MINIATURE CIRCUIT BREAKER
FP	FOUR POLE
TP	THREE POLE
DP	DOUBLE POLE
SP	SINGLE POLE
	EARTH SWITCH
	AMMETER SELECTOR SWITCH
	VOLTMETER SELECTOR SWITCH
	DIGITAL POWER METER
	CURRENT TRANSFORMER
	INDICATION LIGHT

(SHEET 1 OF 1)

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)



WATER AND SANITATION
AGENCY, LAHORE



A Joint Venture

04		
03		
02		
01		
REV.	DATE	DESCRIPTION

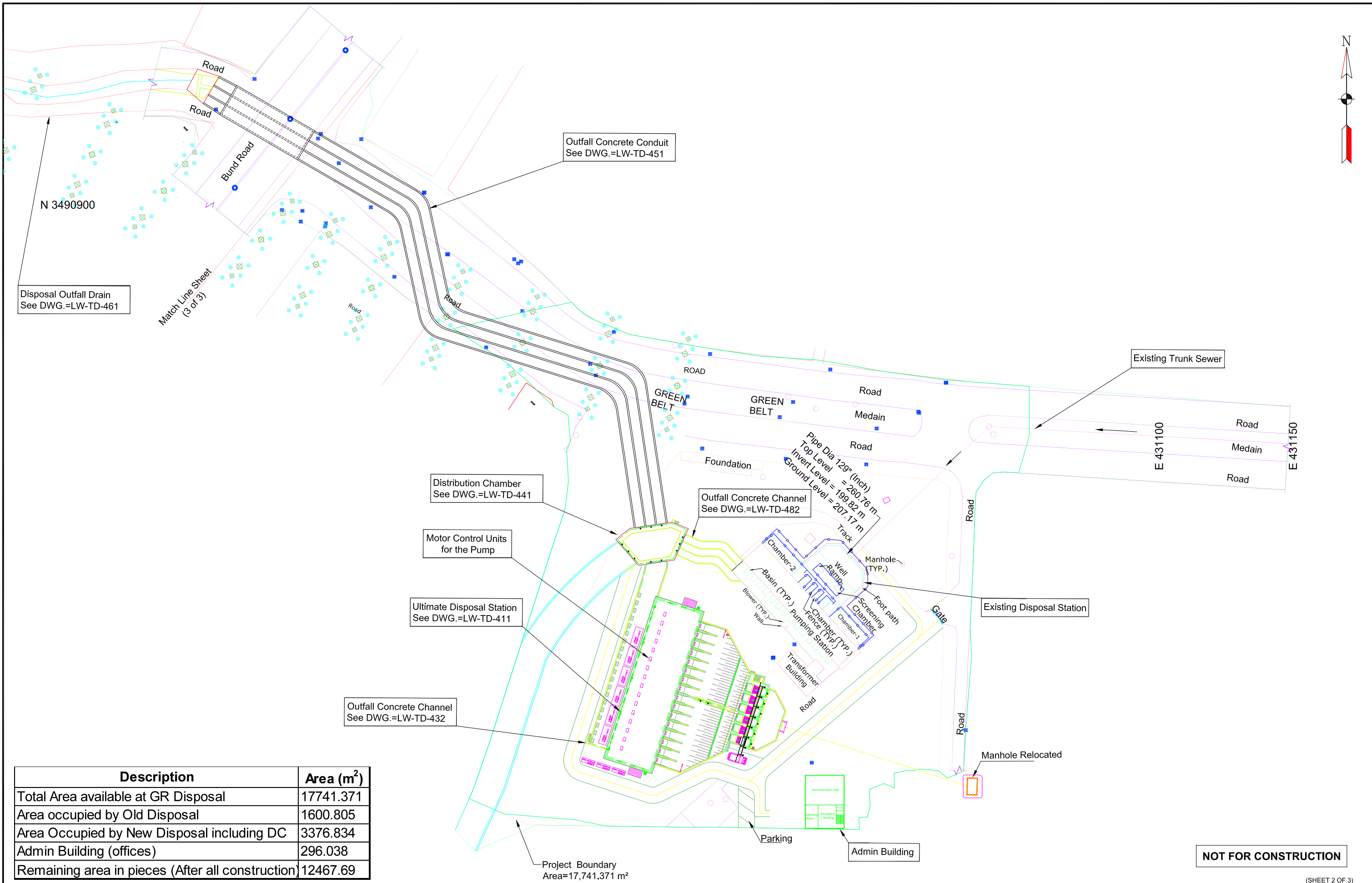
DRAWN	MUHAMMAD ARSLAN
CHECKED BY	SHAHZAIB SHABBIR
RECOMMENDED	
CHD./VER.	
APPROVED	

VENTILATION SYSTEM (SINGLE LINE DIAGRAM) TYPICAL

DATE:-
SEPTEMBER, 2024

DRAWING NO.:-
LW-TD-153

附录-B



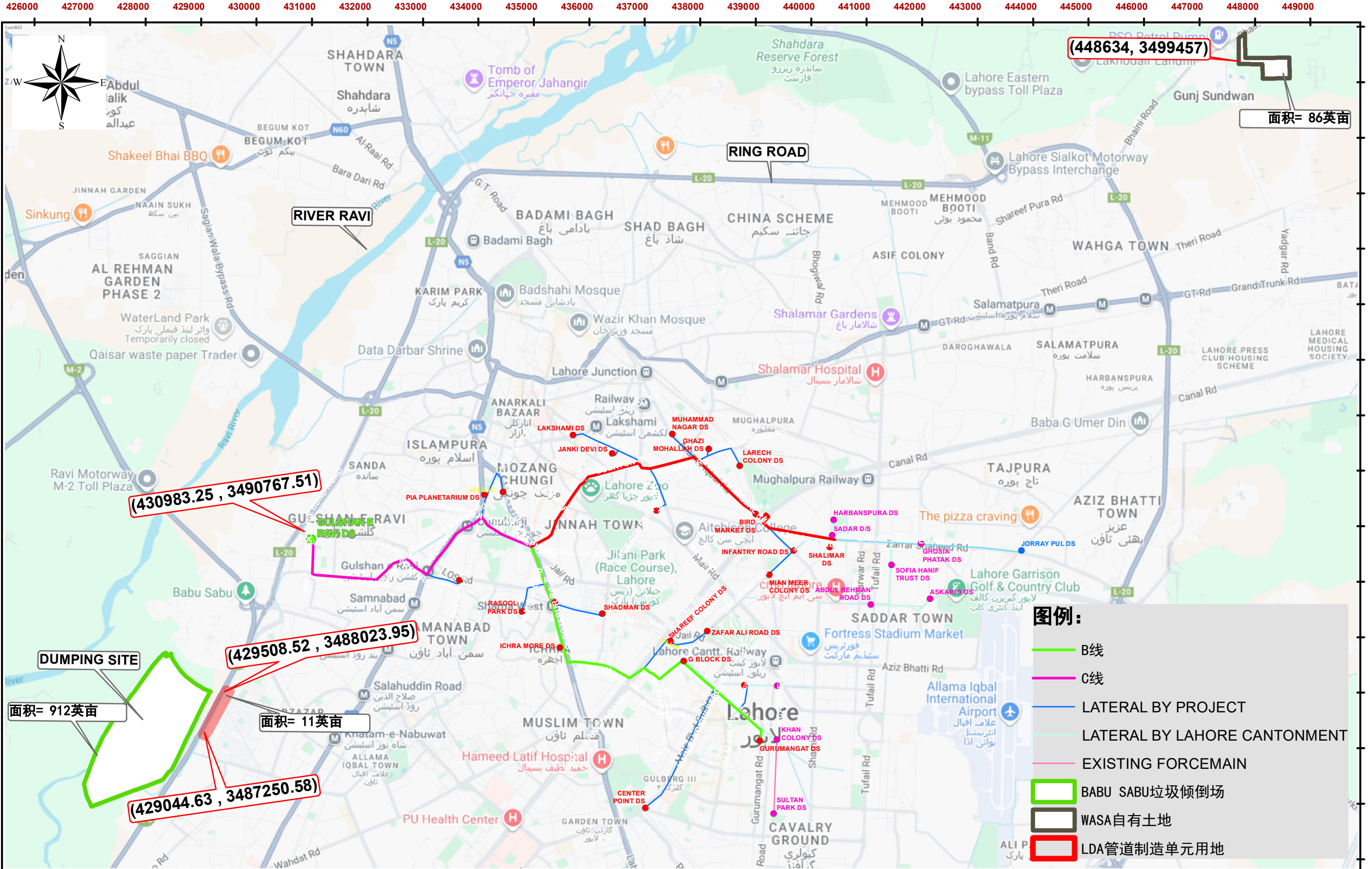
LAHORE WASTEWATER AND DRAINAGE MANAGEMENT PROJECT (LWDMP) - SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E-RABI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY)

 WATER AND SANITATION AGENCY, LAHORE	    <i>A Joint Venture</i>	04 03 02 01 REV. DATE DESCRIPTION	DRAWN MUHAMMAD HAMMAD NASEER CHECKED BY MUHAMMAD SAQIB RAZA RECOMMENDED CHD./VER. APPROVED	ULTIMATE DISPOSAL STATION MASTER LAYOUT PLAN PUMP HOUSE	
				DATE:-	NOVEMBER, 2025
				DRAWING NO.:-	LW - TD - 402

附录-C

序号	提升泵站名称	泵的数量与容量 (电机额定功率)	变压器 (KVA)	发电机 (KVA)	备注
1	Mian Mir	1台x 4立方英尺/秒 (60HP)	100	不可用	
2	Infantry Road	1台x 4立方英尺/秒 (50HP) 1台x 4立方英尺/秒 (50HP) 1台x 6立方英尺/秒 (50HP) 1台x 10立方英尺/秒 (75HP)	200	200	根据PMC设计需要迁移发电机
3	Bird Market	1台x 10立方英尺/秒 (100HP) 1台x 10立方英尺/秒 (100HP) 1台x 25立方英尺/秒 (200HP)	400	300	根据PMC设计需要迁移发电机
4	Larech Colony	1台x 2立方英尺/秒 (20HP) 1台x 6立方英尺/秒 (60HP)	100	200	根据PMC设计需要迁移发电机
5	Ghazi Mohalla	1台x 4立方英尺/秒 (30HP) 1台x 6立方英尺/秒 (60HP)	100	200	根据PMC设计需要迁移发电机
6	Muhammad Nagar	1台x 6立方英尺/秒 (80HP)	50	200	
7	Governor House	1台x 6立方英尺/秒 (60HP) 1台x 6立方英尺/秒 (60HP) 1台x 6立方英尺/秒 (60HP)	2 x 200KVA	不可用	
8	Lakshmi Chowk	1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP) 1台x 25立方英尺/秒 (215HP)	400 & 1000KVA	500KVA + 500KVA + 1000KVA	根据PMC设计需要迁移发电机
9	Lytton Road	1 No x 6Cusec (60HP) 1 No x 10Cusec (80HP) 1 No x 6Cusec (40HP)	200KVA	200KVA	根据PMC设计需要迁移发电机
10	PIA Planterium	1 No x 25Cusec (215HP) 1 No x 12Cusec (100HP) 1 No x 12Cusec (100HP) 1 No x 12Cusec (100HP) 1 No x 25Cusec (215HP)	400 & 200KVA	200KVA & 1000KVA	
11	Gurumangat Road	1 No x 25Cusec (150HP) 1 No x 15Cusec (150HP) 1 No x 15Cusec (150HP)	200KVA	500KVA	
12	Q Block Gulberg	1 No x 25Cusec (150HP) 1 No x 6Cusec (50HP)	200KVA	200KVA	
13	G Block Gulberg	1 No x 25Cusec (150HP) 1 No x 25Cusec (150HP) 1 No x 25Cusec (150HP) 1 No x 25Cusec (150HP)	1000KVA	500KVA	根据PMC设计需要迁移发电机
14	Zafar Ali Road	1 No x 6Cusec (50HP) 1 No x 6Cusec (40HP)	100KVA	200KVA	根据PMC设计需要迁移发电机
15	Shareef Colony	1 No x 6Cusec (40HP) 1 No x 6Cusec (40HP)	50KVA	200KVA	根据PMC设计需要迁移发电机
16	Shadman Colony	1 No x 6Cusec (40HP) 1 No x 25Cusec (150HP) 1 No x 25Cusec (150HP)	2 x 630KVA	630KVA	
17	Shama	1 No x 10Cusec (75HP) 1 No x 6Cusec (80HP) 1 No x 6Cusec (60HP)	100KVA	200KVA	根据PMC设计需要迁移发电机
18	Ichra	1 No x 6Cusec (60HP) 1 No x 6Cusec (80HP) 1 No x 6Cusec (80HP)	800KVA	300KVA	
19	Rasool Park	1 No x 25Cusec (150HP) 1 No x 15Cusec (110HP) 1 No x 08Cusec (80HP) 1 No x 06Cusec (60HP) 1 No x 06Cusec (110HP)	200KVA & 630KVA	500KVA	根据PMC设计需要迁移发电机
20	Poonch Road	1 No x 10Cusec (80HP) 1 No x 10Cusec (80HP) 1 No x 20Cusec (150HP)	400KVA	400KVA	根据PMC设计需要迁移发电机
21	Jhan ki Devi	1无x 01立方英尺/秒 (20马力)	50KVA	不可用	
22	Centre Point	1台x 08立方英尺/秒 (50HP) 1台x 06立方英尺/秒 (50HP) 1台x 06立方英尺/秒 (60HP)	400KVA	300KVA	根据PMC设计需要迁移发电机
23	Harbanspura /Karachi Phatak处置	5台x 25立方英尺/秒 (150HP)	1000KVA	1000KVA	在建 (预计于2026年第一季度完工)

附录-D



LAHORE污水处理与排水管理项目 (LWDMP) - 从LARECHS COLONY到GULSHAN-E-RAVI, LAHORE的污水系统（采用非开挖技术）														
 水与卫生 机构, LAHORE	    A Joint Venture	REVISION	04			CONTROL	绘制		WASA自有土地		比例 1:70,000			
			03				校核							
			02				推荐							
			01				CHD./VER.		日期 :-			2025年12月	LW-TD-005A	版次:-
			版次	日期	说明		批准							